VTPEH 6270 - Check Point 04

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1. Objective and Approach

1.1 Specific Objective

Is depression score associated with daily carbohydrate intake?

1.2 Data Description

```
# Load data
setwd("/Users/shiyi11/Desktop/R/6270/6270 Check Point 04")
nhdc = read.csv("nhanes_L_diet_depression.csv")

# create data description table
data_description = data.frame(
  Variable_Name = c("Depression_Score", "DR1TCARB", "RIAGENDR"),
  Variable_Meaning = c(
    "Depression symptom score",
    "Daily carbohydrate intake(g)",
    "Participant gender"
  ),
  Variable_Type = c(
    "Continuous",
    "Continuous",
    "Categorical"
```

```r
  ),
  Variable_Class_in_R = c(
    class(nhdc$Depression_Score),
    class(nhdc$DR1TCARB),
    class(nhdc$RIAGENDR)
  )
)
```

```r
# display table
kable(
  data_description,
   col.names = c(
    "Variable name",
    "Meaning",
    "Type in dataset",
    "Class in R"
  ),
  caption = "Description of variables used in the research question"
)
```

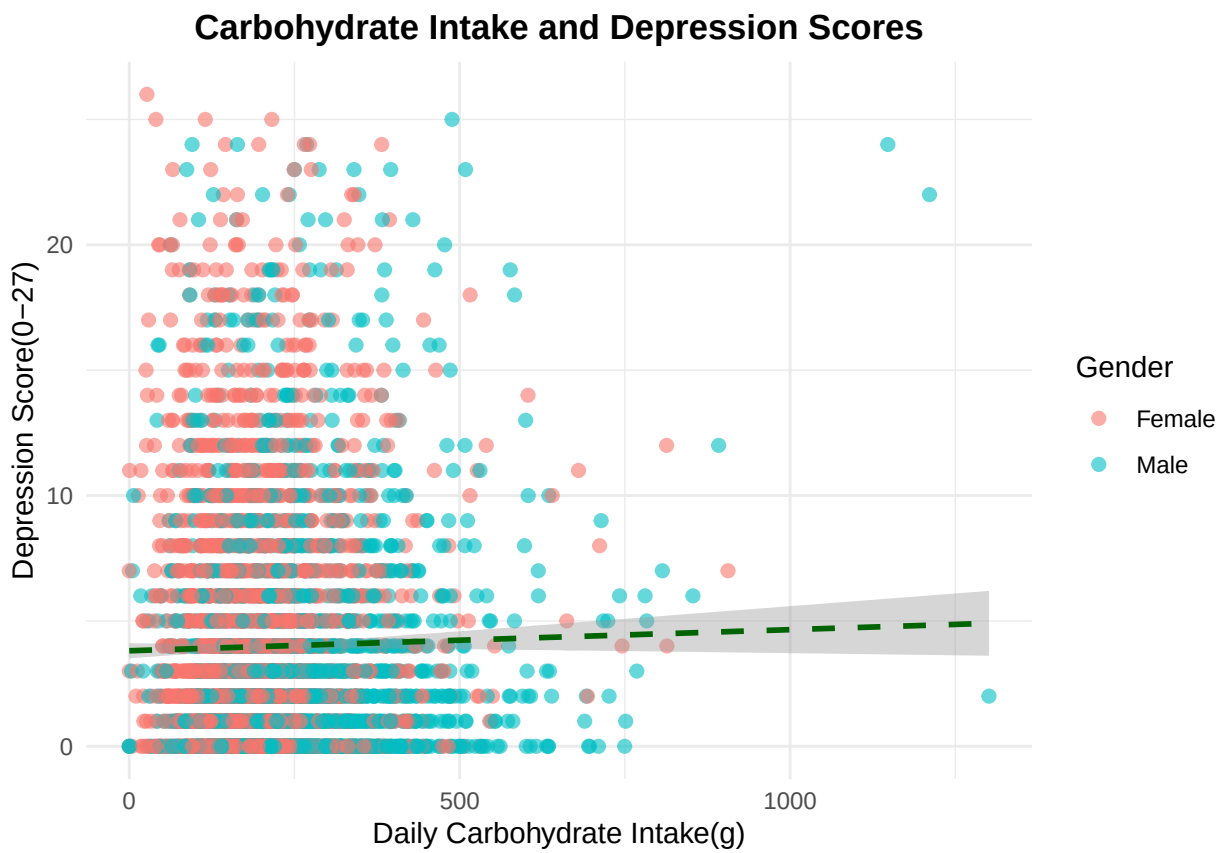
Table 1: Description of variables used in the research question

Variable name	Meaning	Type in dataset	Class in R
Depression_Score	Depression symptom score	Continuous	integer
DR1TCARB	Daily carbohydrate intake(g)	Continuous	numeric
RIAGENDR	Participant gender	Categorical	integer

1.3 Data Visualization

```r
ggplot(nhdc,
       aes(x = DR1TCARB,
           y = Depression_Score,
           color = Gender)) +
  geom_point(alpha = 0.6, size = 2) +

# Add a linear regression line to visualize the average association
  geom_smooth(method = "lm",
              linetype = "dashed",
              color = "darkgreen",
              se = TRUE) +
  labs(
    title = "Carbohydrate Intake and Depression Scores",
    x = "Daily Carbohydrate Intake(g)",
    y = "Depression Score(0-27)",
    color = "Gender"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
```



I choose the scatter plot to visualize the data. As is shown in the generated plot, each point represents an individual and colors indicate gender. The dashed line shows the fitted linear regression trend and the grey ribbon represents the 95% confidence interval.

1.4 Description of Plausible Relationships

Based on the visual inspection of the scatter plot, daily carbohydrate intake and depression score appear to follow a weak positive linear association. The regression line suggests that higher carbohydrate intake may be associated with slightly higher depression scores. However, the large variability in the data indicates that this association is relatively weak. In addition, the distribution of points for males and females appears largely overlapping, indicating that the relationship between carbohydrate intake and depression score does not differ substantially by gender.

1.5 Parameters of Interest

```
parameters_table = data.frame(
  Parameter = c(
    "Intercept",
    "Slope for carbohydrate intake",
    "Noise term"
  ),
  Symbol = c(
    "b",
```

```
    "a",
    "e"
  ),
  Description = c(
    "Expected depression score when carbohydrate intake = 0",
    "Change in depression score associated with a 1g increase in carbohydrate intake",
    "Unexplained random variability"
  )
)
```

```
kable(
  parameters_table,
  col.names = c("Parameter", "Symbol", "Description"),
  caption = "Parameters of interest for the proposed relationship"
)
```

Table 2: Parameters of interest for the proposed relationship

Parameter	Symbol	Description
Intercept	b	Expected depression score when carbohydrate intake = 0
Slope for carbohydrate intake	a	Change in depression score associated with a 1g increase in carbohydrate intake
Noise term	e	Unexplained random variability

The mathematical relationship I use is: Y=b+aX+e, and in my situation, Y means the Depression Score, X is the Daily Carbohydrate Intake.

2. Simulation

2.1 Simulation Basis

```
# Set the seed for reproducibility
set.seed(123)

# Number of observations
nn = 1000

# Simulate carbohydrate intake
carb_intake = rnorm(nn, mean = 250, sd = 120)
carb_intake = pmax(0, carb_intake)

# Define model parameters
intercept = 3
slope_carb = 0.002

# Random noise
epsilon = rnorm(nn, mean = 0, sd = 4)
```

```r
# Simulate depression score
depression_score = intercept +
                    slope_carb * carb_intake +
                    epsilon

# Restrict simulated depression scores to the realistic PHQ-9 range from 0 to 27
depression_score = round(pmax(0, pmin(27, depression_score)))

# Create simulated dataset
data = data.frame(
  carbohydrate_intake = carb_intake,
  depression_score = depression_score
)

# Display the first few rows of the simulated dataset
head(data)
```

2.2 Simulation Output

```r
# Summary table
summary_table = data.frame(
  Variable = c("Carbohydrate intake", "Depression score"),
  Mean = c(
    mean(data$carbohydrate_intake),
    mean(data$depression_score)
  ),
  SD = c(
    sd(data$carbohydrate_intake),
    sd(data$depression_score)
  ),
  Q1 = c(
    quantile(data$carbohydrate_intake, 0.25),
    quantile(data$depression_score, 0.25)
  ),
  Median = c(
    median(data$carbohydrate_intake),
    median(data$depression_score)
  ),
  Q3 = c(
    quantile(data$carbohydrate_intake, 0.75),
    quantile(data$depression_score, 0.75)
  )
)

summary_table
```

```
##               Variable     Mean         SD       Q1   Median       Q3
## 1 Carbohydrate intake 252.4096 117.877590 174.6011 251.1052 329.7522
## 2    Depression score   4.0810   3.446493   1.0000   4.0000   7.0000
```

2.3 Simulation Function

```
f_sim_nhdc = function(effect_size,
                      noise_sd,
                      sample_size){

  # Carbohydrate intake vector
  v_carb = rnorm(sample_size,
                 mean = 250,
                 sd = 120)
  v_carb = pmax(0, v_carb)

  # Model parameters
  intercept = 3

  # Generate random noise
  epsilon = rnorm(sample_size,
                  mean = 0,
                  sd = noise_sd)

  # Depression score vector
  v_depression = intercept + effect_size * v_carb + epsilon

  # Restrict depression score between 0 and 27
  v_depression = round(pmax(0, pmin(27, v_depression)))

  # Combine data
  data_sim = data.frame(
    carbohydrate_intake = v_carb,
    depression_score = v_depression
  )

  return(data_sim)
}
```

2.4 Simulation Automation

```
# Initialize lists to store results
list_datasets = list()
df_stats = data.frame(
  run_id = NA,
  sim = NA,
  effect_size = NA,
  sample_size = NA,
  noise_sd = NA,
  carb_mean = NA,
  carb_sd = NA,
  carb_q1 = NA,
  carb_median = NA,
  carb_q3 = NA,
  dep_mean = NA,
```

```r
  dep_sd = NA,
  dep_q1 = NA,
  dep_median = NA,
  dep_q3 = NA,
  slope = NA
)
sim_index = 1

# Run simulations across all scenarios
for (aa in seq(0, 0.009, 0.001)) {
# loop on effect size
  for (nn in c(50, 100, 150, 200, 300, 400, 500, 700, 900, 1200)) {
# loop on sample size
    for (ee in c(2, 4, 6)) {
# loop on noise levels
    for (ii in 1:10) {
# Run simulation 10 times
      # Generate data
      sim_data = f_sim_nhdc(
        effect_size = aa,
        noise_sd = ee,
        sample_size = nn
      )

      # Store full dataset
      list_datasets[[sim_index]] = sim_data

      # Store summary statistics
      df_stats[sim_index, ] <- NA
      df_stats[sim_index, "run_id"] = sim_index
      df_stats[sim_index, "effect_size"] = aa
      df_stats[sim_index, "sim"] = ii
      df_stats[sim_index, "sample_size"] = nn
      df_stats[sim_index, "noise_sd"] = ee

# Store summary statistics for carbohydrate intake
      df_stats$carb_mean[sim_index] =
        round(mean(sim_data$carbohydrate_intake), 2)
      df_stats$carb_sd[sim_index] =
        round(sd(sim_data$carbohydrate_intake), 2)
      df_stats$carb_q1[sim_index] =
        round(quantile(sim_data$carbohydrate_intake, 0.25), 2)
      df_stats$carb_median[sim_index] =
        round(median(sim_data$carbohydrate_intake), 2)
      df_stats$carb_q3[sim_index] =
        round(quantile(sim_data$carbohydrate_intake, 0.75), 2)

# Store summary statistics for depression score
      df_stats$dep_mean[sim_index] =
        round(mean(sim_data$depression_score), 2)
      df_stats$dep_sd[sim_index] =
        round(sd(sim_data$depression_score), 2)
      df_stats$dep_q1[sim_index] =
```

```r
        round(quantile(sim_data$depression_score, 0.25), 2)
      df_stats$dep_median[sim_index] =
        round(median(sim_data$depression_score), 2)
      df_stats$dep_q3[sim_index] =
        round(quantile(sim_data$depression_score, 0.75), 2)

# Fit linear model for each simulated dataset
model = lm(depression_score ~ carbohydrate_intake, data = sim_data)

# Store estimated slope
df_stats[sim_index, "slope"] = coef(model)[2]

      sim_index = sim_index + 1
      }
    }
  }
}

# Check summary statistics
head(df_stats)
```

```
##   run_id sim effect_size sample_size noise_sd carb_mean carb_sd carb_q1
## 1      1   1           0          50        2    250.62  107.53  177.80
## 2      2   2           0          50        2    229.54  100.68  172.81
## 3      3   3           0          50        2    250.75  115.35  172.92
## 4      4   4           0          50        2    247.97  112.32  166.56
## 5      5   5           0          50        2    223.94  121.37  156.64
## 6      6   6           0          50        2    250.30  121.11  172.92
##   carb_median carb_q3 dep_mean dep_sd dep_q1 dep_median dep_q3         slope
## 1      243.47  326.37     2.56   1.58      1        3.0   3.75  0.0007367815
## 2      231.51  292.15     2.84   1.57      2        3.0   4.00 -0.0003542495
## 3      244.74  329.81     2.64   1.60      1        3.0   4.00 -0.0004258456
## 4      249.89  322.89     3.42   1.99      2        3.5   5.00  0.0020117598
## 5      215.70  278.34     3.30   1.87      2        3.0   4.00  0.0001485459
## 6      256.74  340.44     3.40   1.81      2        4.0   5.00 -0.0042261282
```

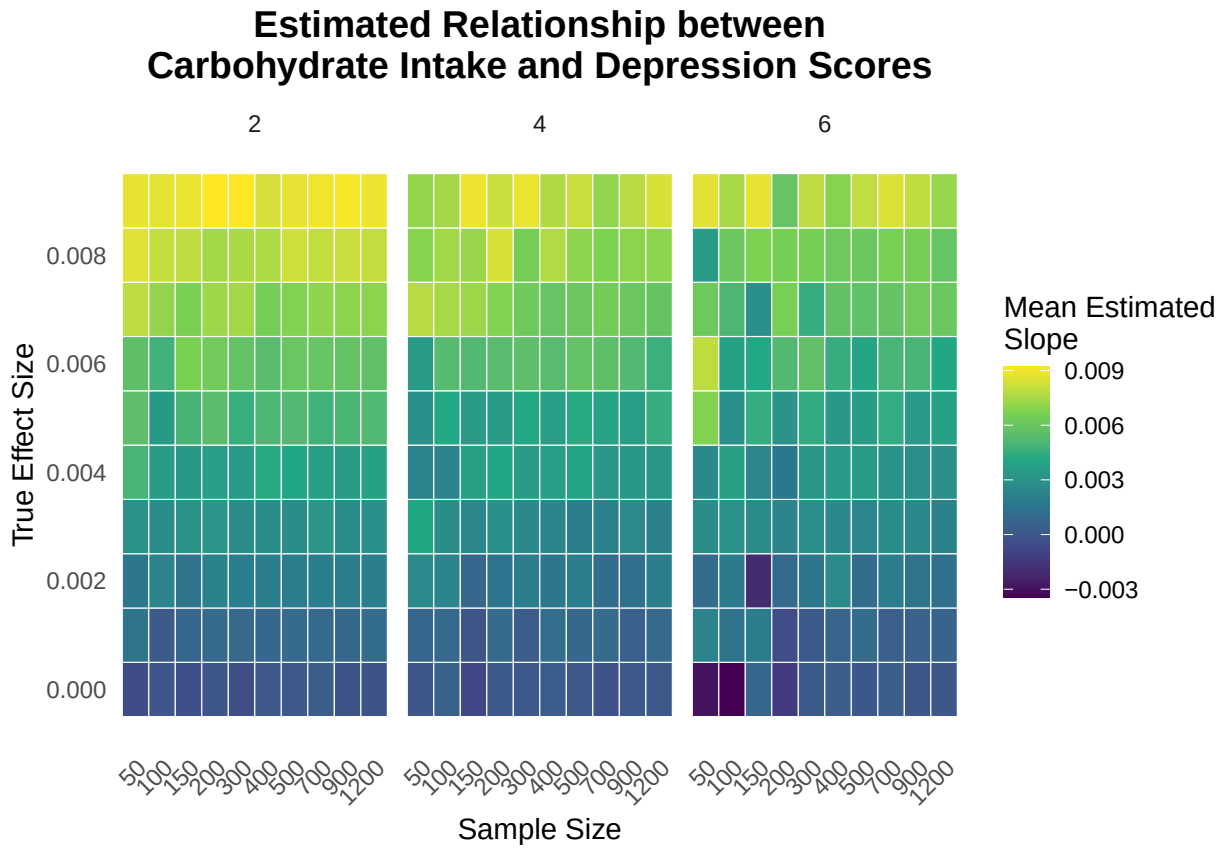
3. Visualization

```r
# Compute the average estimated slope across repeated simulations
df_plot = df_stats %>%
  group_by(effect_size, sample_size, noise_sd) %>%
  summarise(
    mean_slope = mean(slope),
    .groups = "drop"
  )

# Convert sample size to a factor to control the order on the x-axis
df_plot$sample_size = factor(
  df_plot$sample_size,
  levels = c(50, 100, 150, 200, 300, 400, 500, 700, 900, 1200)
```

```r
)

# Create heatmap visualizing the estimated relationship between carbohydrate intake and depression scor
ggplot(df_plot,
       aes(x = sample_size,
           y = effect_size,
           fill = mean_slope)) +
  geom_tile(color = "white") +
  facet_wrap(~noise_sd, nrow = 1) +
  scale_fill_viridis_c() +
  scale_y_continuous(
    breaks = seq(0, 0.009, 0.002),
    labels = scales::number_format(accuracy = 0.001)
  ) +
  labs(
    title = "Estimated Relationship between\nCarbohydrate Intake and Depression Scores",
    x = "Sample Size",
    y = "True Effect Size",
    fill = "Mean Estimated\nSlope"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 14
    ),
# Rotate x-axis labels to avoid overlap
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    ),
    panel.grid = element_blank()
  )
```



4. Interpretation

The heatmap shows that the estimated relationship between carbohydrate intake and depression score varies across the simulated parameter settings.

Overall, the estimated slope increases as the true effect size increases. When the true effect size is close to zero, the estimated slopes remain close to zero across most sample sizes, indicating little or no detectable association between carbohydrate intake and depression scores. When the true effect size is very small, some simulations produce slightly negative slope estimates. Noise levels also influence the stability of the estimated relationship. At higher noise levels (noise = 6), the estimated slopes appear more variable and less consistent across sample sizes, suggesting that random variation can obscure the underlying association. In contrast, when noise is lower (noise = 2), the estimated slopes more closely follow the true effect sizes. As for the sample size, larger sample sizes tend to produce more stable estimates of the slope. As sample size increases, the estimated slopes become more consistent across simulations, indicating that larger datasets improve the reliability of detecting the relationship between carbohydrate intake and depression scores.

5. AI Use Disclosure Statement

ChatGPT was used to draft some of the rough code, explain R functions and assist with improving and debugging R code. All analysis, interpretations and final written answer were reviewed and completed by the author.